Financial Engineering Mid-Review Presentation Guide (INR)

Capital Budgeting and Infrastructure Valuation with Monte Carlo Risk Simulation (INR)

Project Title: Capital Budgeting and Infrastructure Project Valuation with Monte Carlo Risk Simulation	**Dataset:** Kaggle / World Bank PPI Database
Currency Standard: Indian Rupees (INR ■ Crores & ■/kWh)	**Target Audience:** CFO / Infrastructure Committee

1. Project Milestones: Mid-Review vs. Final Review

Phase	Key Deliverables & Capabilities	Status
Mid-Review (Today)	• Public Kaggle dataset cleaning and Indian Rupees standardization. • **Analysis 1:** Deterministic Valuation (NPV, IRR, LCOE, PI, Payback in ■ Crores). • **Analysis 2:** 5,000-scenario Monte Carlo Risk Engine & 2D Sensitivity Heatmap. • Deployed live on Streamlit Cloud.	**100% COMPLETED**
Final Review (Future)	• **Multi-Asset Portfolio Optimization:** Balancing Solar vs Wind vs Battery Storage. • **Carbon Offset Revenues:** Integrating SECI / Carbon Credit cash flows. • **GPU High-Throughput Scaling:** 1,000,000+ path CUDA batch rollout on compute node. • **Bank Debt Financing:** Loan amortization & DSCR schedules.	Planned Next Steps

2. Step-by-Step UI Component Breakdown (INR)

UI Component	What It Shows (INR)	Why It Is Useful in Real Life
1. Sidebar Inputs	Project presets, CapEx (■ Cr), OpEx (■ Cr), PPA Tariff (■/kWh), WACC (%), Inflation (%).	Allows real-time tweaking of project parameters matching Indian tariff auctions.
2. Tab 1 Scorecard	**NPV:** +■275.50 Cr | **IRR:** 18.52% **LCOE:** ■1.99/kWh | **PI:** 2.10x **Payback:** 7 Years	Provides immediate executive viability: NPV shows wealth creation, IRR beats 8% WACC by +10.5%, and LCOE proves profit vs ■3.85/kWh tariff.
3. Tab 1 Cash Flow Waterfall	Year 0 outflow (-■250 Cr) + annual positive inflows (+■46.7 Cr/yr) reaching break-even at Year 7.	Visually tracks cumulative debt reduction and profit accumulation over 20 years.
4. Tab 2 Monte Carlo Engine	Histogram of 5,000 simulated scenarios, 95% VaR (+■152.8 Cr), Probability of Loss (0.00%).	Proves the project remains resilient and profitable even under severe tariff and cost shocks.
5. Tab 2 2D Sensitivity Heatmap	Color-coded matrix of NPV (■ Cr) across PPA Tariffs (■2.90 to ■4.80/kWh) vs. WACC (6% to 12%).	Shows the exact boundary where the project shifts from profit to loss.

3. Spoken Presentation Script for Mid-Review (INR)

Phase 1: Introduction & Problem Statement

■ **Action on Screen:** Show Title & Overview on website.

> ***What to Say:*** *"Good morning/afternoon, Sir. Our project is titled Capital Budgeting and Infrastructure Project Valuation with Monte Carlo Risk Simulation. When an Indian energy company or government invests ■250 Crores to build a 100MW Solar Park or Wind Farm, the CFO needs to know: Will 20 years of electricity revenues at ■3.85 per unit justify this massive upfront cost, and what is our financial risk if tariffs drop? We built this quantitative tool to answer that."*

Phase 2: Analysis 1 (Deterministic Valuation in INR)

■ **Action on Screen:** Click on Tab 1, point to metric cards and the Waterfall chart.

> ***What to Say:*** *"In Tab 1, we pull real project benchmarks from our Kaggle / World Bank dataset in Indian Rupees. Discounting all cash flows at an 8% cost of capital (WACC): Our Net Present Value (NPV) is +■275.50 Crores, proving the project is highly value-accretive. The Internal Rate of Return (IRR) is 18.52%, beating our 8% hurdle rate by +10.5%. Our Levelized Cost of Energy (LCOE) is ■1.99 per unit (kWh), giving a strong profit margin against our ■3.85 PPA selling tariff. The Cash Flow Waterfall chart proves our ■250 Cr investment is fully paid back in 7 Years."*

Phase 3: Analysis 2 (Monte Carlo Risk Simulation in INR)

■ **Action on Screen:** Click on Tab 2, drag slider to 10,000 runs, point to histogram and heatmap.

> ***What to Say:*** *"In Tab 2, a static NPV is not enough because real tariffs fluctuate and construction often faces overruns. So we built a 5,000-scenario Monte Carlo Risk Engine. In each run, we simultaneously shock tariffs by ±15%, CapEx overruns up to +25%, and inflation. As you can see, our Probability of Loss is 0.00%, and our 95% Value-at-Risk is +■152.8 Crores. The 2D sensitivity heatmap shows our exact break-even boundary across different interest rates and unit tariffs."*

Phase 4: Conclusion & Next Steps

■ **Action on Screen:** Show Tab 4 memo, then return to Tab 1.

> ***What to Say:*** *"Tab 4 auto-generates an Executive Investment Memo recommending board approval. For our Final Review, our next milestones are: (1) multi-asset portfolio optimization for Solar vs Wind vs Battery Storage, (2) integrating Carbon Offset revenues, and (3) scaling our Monte Carlo engine to 1,000,000 paths on our college GPU cluster. Thank you, Sir!"*